SATELLITE INTERNET

Satellite Internet:
Bridging the Global Digital Divide from Space

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BCA - Semester VI

INTRODUCTION

- The internet is an essential part of modern life
- Most internet services depend on terrestrial infrastructure
- Many regions still lack reliable connectivity
- Satellite Internet provides internet access from space
- It enables global communication without ground cables

EVOLUTION OF THE INTERNET

- Internet began as a research and communication network
- Early access used wired connections like dial-up and DSL
- Broadband and fiber improved speed and reliability
- Mobile networks (3G, 4G, 5G) enabled wireless internet
- Many areas remain unreachable by terrestrial networks

WHY SATELLITE INTERNET IS ESSENTIAL

Vast remote and rural areas still lack basic internet access.

Deploying traditional fiber cables is extremely costly and time-consuming.

Geographical barriers like mountains, oceans, and deserts block ground networks.

Terrestrial infrastructure is vulnerable to natural disasters and disruptions.

Satellite internet offers critical, widespread connectivity from orbit, bypassing these challenges.

WHAT IS SATELLITE INTERNET?

Delivers internet connectivity directly from satellites orbiting Earth, bypassing traditional ground infrastructure.

Works by transmitting data between your device, a satellite, and ground stations, creating a seamless connection.

Eliminates the need for extensive cables or mobile towers, making it accessible where traditional networks can't reach.

Ideal for remote, rural, and underserved areas, bringing high-speed internet to isolated communities.

Provides widespread global coverage, ensuring connectivity even in challenging terrains or during emergencies.

BASICS OF SATELLITE COMMUNICATION

1. Satellite communication uses radio signals to send information.
2. Data travels between ground stations on Earth and satellites orbiting above.
3. Satellites act as relay stations, receiving signals and sending them back to Earth.
4. This enables communication across vast distances and provides global coverage.
5. These core principles form the foundation of satellite internet services.

Components of a Satellite Communication System

User Terminal:

Your equipment (like a dish and modem) that sends and receives satellite data.

Communication Satellite:

The orbiting satellite that captures signals and sends them back to Earth.

Ground Station:

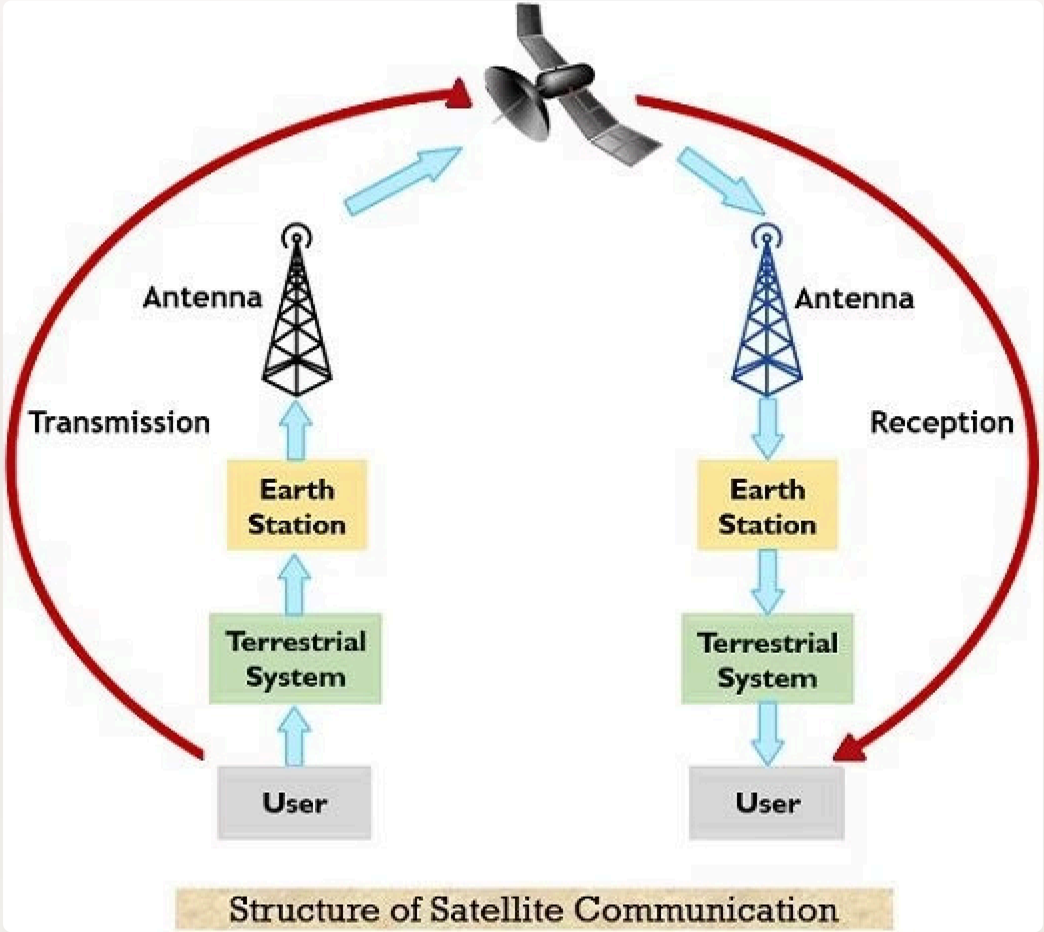
Large antennas on Earth that connect the satellite network to the global internet.

Control Station:

Earth-based centers that manage and monitor the satellite's operations.

Network Infrastructure:

The underlying system that routes data between ground stations and the wider internet.

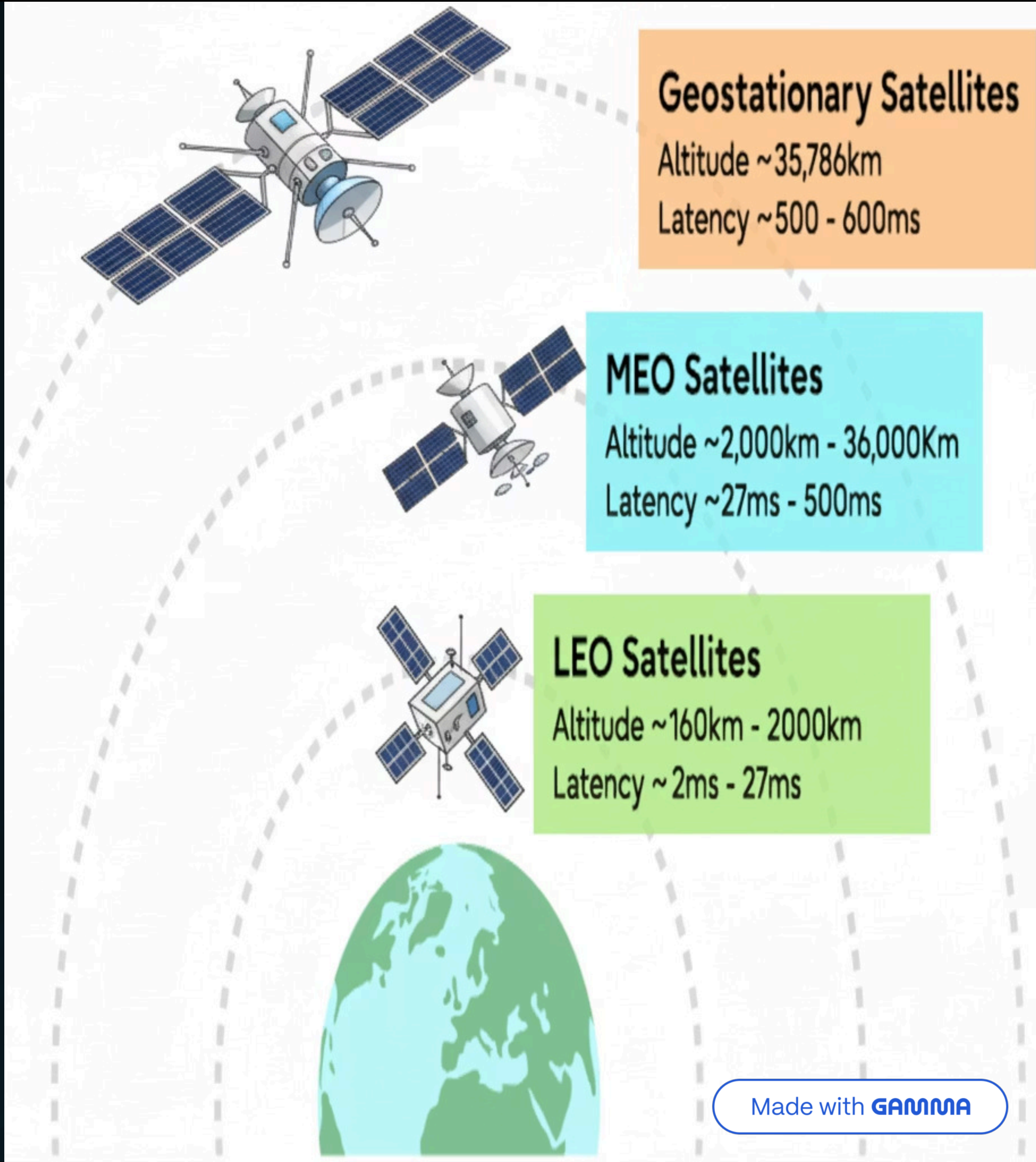


SATELLITE COMMUNICATION PROCESS

1. User generates data (voice, video, or internet request)
2. Data travels through the terrestrial system to the Earth station
3. Earth station transmits the signal to the satellite (**uplink**)
4. Satellite receives, amplifies, and retransmits the signal
5. Receiving Earth station gets the signal (**downlink**)
6. Terrestrial system delivers data to the destination user

SATELLITE ORBITS

1. Satellites travel in specific paths around Earth, known as orbits.
2. The chosen orbit impacts factors like signal coverage, speed, and latency (signal delay).
3. Different types of orbits are optimized for various communication purposes.
4. For satellite internet, three primary orbit types are used: Low Earth Orbit (LEO), Medium Earth Orbit (MEO), and Geostationary Earth Orbit (GEO).



LOW EARTH ORBIT (LEO)

Very Close to Earth

LEO satellites orbit very close to our planet, typically between 500 and 2000 kilometers up. This closeness helps signals travel better and makes communication more efficient and powerful.

Super Low Delay (Latency)

Because they are closer, LEO satellites greatly shorten how far signals need to go. This leads to very low signal delay (called latency) and super-fast internet speeds. It's perfect for things that need instant responses.

Great for Real-Time Use

The naturally low latency of LEO systems makes them ideal for things that need instant replies. This includes smooth video calls, online gaming, live streaming, and important operations where even tiny moments count.

Many Satellites for Global Coverage

Since LEO satellites are closer to Earth, each one covers only a small area. To get continuous internet around the world, many LEO satellites (a "constellation") must work together.

Leading Modern Satellite Internet

LEO technology is changing satellite internet. It offers much lower delay and faster speeds than older systems. LEO satellites are now bringing fast internet to remote places everywhere.

MEDIUM EARTH ORBIT (MEO)

Middle Altitude

MEO satellites orbit between 2,000 and 35,786 kilometers, balancing between LEO and GEO orbits.

Balanced Performance

Offers a good mix of signal speed and coverage area; faster than GEO, covers more ground than LEO.

Fewer Satellites

Higher orbits mean each satellite covers a larger area, requiring fewer for global coverage compared to LEO.

Key Applications

Primarily used for navigation systems like GPS, Galileo, and GLONASS, and for regional internet services.

GEOSTATIONARY EARTH ORBIT (GEO)

Very High Up

GEO satellites are placed very far from Earth, about 36,000 km high. They orbit right above the equator.

Always in Sight

These satellites move at the exact same speed as Earth rotates. This means they appear to stay in the same spot in the sky, always watching over one area.

Huge Coverage Area

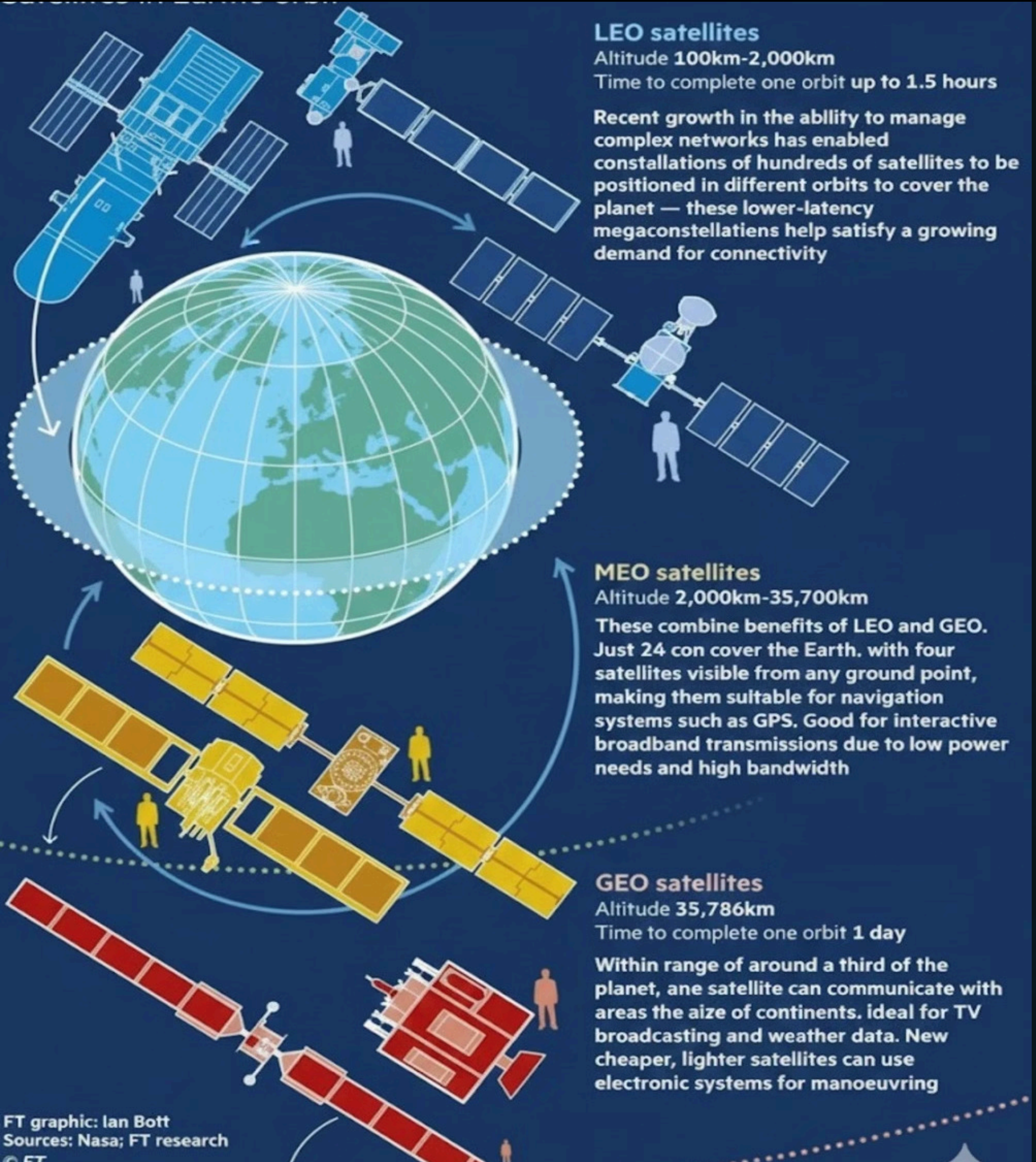
One GEO satellite can cover a very large part of Earth, sometimes an entire continent. This is great for broadcasting TV and radio to many people.

Fewer Satellites Needed

Because each satellite covers so much ground, you don't need many of them to provide coverage across the globe. This makes the system simpler to set up.

Long Signal Delay

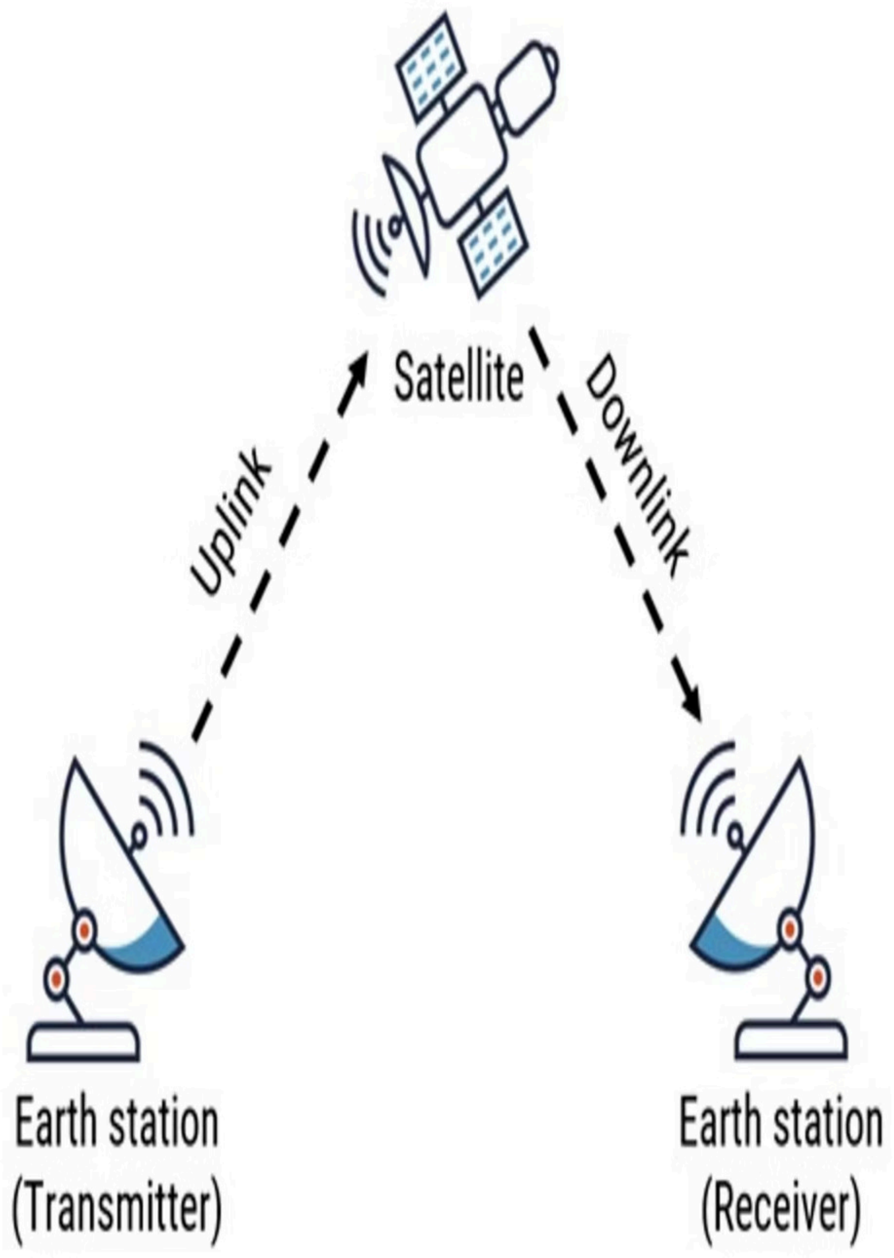
Since GEO satellites are so far away, signals take longer to travel to and from Earth. This causes a noticeable time delay, or "latency," which can affect live calls or fast internet.



Orbit	Altitude	Latency	Coverage
LEO	Low	Low	Small
MEO	Medium	Medium	Medium
GEO	High	High	Large

HOW SATELLITE INTERNET WORKS

1. When you do something online, like click a link or send an email, your request first goes to your satellite dish.
2. Your dish then sends this request, as radio waves, up through the air to a satellite in space.
3. The satellite receives the signal and sends the data back down to a ground station on Earth.
4. This ground station connects to the internet and finds the information you asked for.
5. Finally, the information travels back: from the internet to the ground station, up to the satellite, and then down to your dish. Now you can see it on your computer or phone.



UPLINK AND DOWNLINK COMMUNICATION

- **Uplink:** Your satellite dish sends data (radio waves) from your device to an orbiting satellite. This is the "upward journey" for your online requests.
- **Downlink:** The satellite transmits requested internet data back to your dish. This is the "return path" that brings content to your device.
- **Frequency Separation:** Different frequency bands are used for uplink and downlink to avoid interference and ensure smooth, reliable communication.

FREQUENCY BANDS USED

- Satellite communication uses microwave frequency bands
- **C-Band:** Reliable, less affected by weather
- **Ku-Band:** Used for TV, aviation, and early internet
- **Ka-Band:** High-speed modern satellite internet
- **V-Band:** Future ultra–high bandwidth systems**Comparison Table: Satellite Frequency Bands**

Frequency Band	Frequency Range	Key Features	Common Uses
C-Band	4–8 GHz	Highly reliable, less affected by rain	TV broadcasting, legacy satellite communication
Ku-Band	12–18 GHz	Medium bandwidth, moderate rain fade	DTH services, aviation, maritime internet
Ka-Band	26–40 GHz	High bandwidth, supports high speed	Modern satellite internet (Starlink, OneWeb)
V-Band	40–75 GHz	Very high capacity, future use	Next-generation satellite systems

THE EVOLUTION OF MODERN SATELLITE INTERNET SYSTEMS

- Older GEO satellites orbited far from Earth, leading to high latency and slower internet.
- Modern systems use LEO satellites, which fly much closer (under 1,200 miles).
- This closer proximity allows for significantly faster internet, lower latency, and higher speeds.
- LEO satellites operate in large constellations, providing continuous and global coverage.
- These advancements enable seamless real-time applications like video calls and online gaming, even in remote areas.

Key Players in Satellite Internet

- **Starlink:** SpaceX's LEO constellation provides high-speed internet, focusing on remote and underserved areas.
- **OneWeb:** A global LEO network connecting businesses, governments, and critical sectors like aviation and maritime.
- **Amazon Kuiper:** Amazon's LEO satellite system aims to deliver fast, affordable broadband to homes and businesses, especially in unserved regions.
- Together, these systems are expanding access to fast, affordable internet globally.

APPLICATIONS OF SATELLITE INTERNET

- **Connecting Rural Areas:** Provides internet where traditional infrastructure is unavailable.
- **Aviation & Maritime:** Ensures global communication for air and sea travel.
- **Disaster Relief:** Acts as essential backup when terrestrial networks fail.
- **Military Communications:** Offers secure and resilient defense connectivity.
- **Remote Research & IoT:** Facilitates data transfer in isolated environments.

ADVANTAGES OF SATELLITE INTERNET

- **Global Coverage:** Provides internet in remote or underserved areas.
- **Geographic Flexibility:** Operates effectively in challenging terrains like mountains or oceans.
- **Rapid Deployment:** Quick setup for temporary sites, emergencies, or rapid expansion.
- **Disaster Resilience:** Maintains connectivity during emergencies when terrestrial networks fail.
- **Supports Emerging Tech:** Facilitates remote sensors, IoT, and scientific research.

CHALLENGES OF SATELLITE INTERNET

- **Weather Vulnerability:** Heavy rain, snow, or dense clouds can interrupt or block signals, leading to service outages.
- **Higher Costs:** Specialized equipment and satellite maintenance result in more expensive subscription fees.
- **Space Debris & Congestion:** Increased risk of collisions in orbit due to growing space debris and more satellites.
- **Security & Privacy:** Signals broadcast over wide areas introduce unique security challenges for data protection.

FUTURE OF SATELLITE INTERNET

- **Direct-to-Mobile & 5G/6G Integration:** Unified high-speed access to mobile devices.
- **AI-Optimized Networks:** Smarter traffic management and enhanced reliability.
- **Ultra-Low Latency (LEO):** Responsive internet comparable to terrestrial broadband.
- **Global, Affordable Access:** Bridging the digital divide worldwide.

CONCLUSION

- **Global Connectivity:** Satellite internet offers global connectivity, reaching remote and underserved areas.
- **Overcoming Limits:** It overcomes terrestrial network challenges like cost and inaccessibility, promoting digital inclusion.
- **High Speed & Low Latency:** LEO satellite systems deliver high speed and low latency, comparable to terrestrial broadband.
- **Emergency & Remote Use:** Essential for remote operations, disaster relief, and backup communications where infrastructure is lacking.
- **Future of Internet Access:** A scalable, resilient, and ubiquitous solution defining the future of global internet access.

THANK YOU